

PADM Progress Sheet

Name: Luke Decker

Advisor: Dr. Ting Liu

Project: Parkinson's Analysis Through Digital Motor Metrics (PADM)

Research Period: February 2026- May 2026

Project Overview

Parkinson's Analysis Through Digital Motor Metrics, or PADM, is a non-diagnostic research tool designed to explore early Parkinson's-related patterns through digital motor and cognitive testing. The project focuses on collecting measurable interaction data from keyboard, cursor, touchpad, tracing, and cognitive tasks. The goal is to provide comparative, data-driven insight rather than a medical diagnosis.

The intended program pipeline is:

Input → Motor Tests → Feature Extraction → Model / Similarity Analysis → Output Through UI

The system is designed to collect participant data, structure results, compare patterns against research-based baselines where possible, and display results through a clean participant-facing interface.

Phase 0–1: Foundation and Research Period

Timeframe: Ongoing throughout the project

Completed Objectives

- Defined the primary project deliverables.
- Identified key measurable biometrics:
 - Typing speed
 - Keystroke dynamics
 - Tremor-related inconsistencies
 - Cursor and mouse movement patterns

- Defined the overall program pipeline:
 - Input
 - Motor tests
 - Feature extraction
 - Visual model structure
 - Similarity analysis
 - Non-diagnostic output through the UI
 - Defined initial UI preferences:
 - Menu-based interface
 - Testing screens close or minimize distractions during tasks
 - Results screens reopen after testing
 - Graphs, similarity outputs, tracing results, and general results display are shown after completion
-

Phase 2: Motor Skills and Interface Development

Timeframe: March 7 – March 27, 2026

Completed Objectives as of March 22, 2026

- Implemented rough motor-skills tests.
- Added input capture for:
 - Timing measurements
 - Movement tracking
- Built initial testing components so they could later be connected to the data pipeline and program interface.

Notes

- Fixed or began addressing the NumPy interpreter issue.
- Confirmed Python file path:

C:\Users\decke\AppData\Local\Programs\Python\Python310\python.exe

Phase 2 Continued: Cognitive and Integration Development

Timeframe: March 28 – April 7, 2026

Completed Objectives as of April 7, 2026

- Implemented the cognitive testing framework:
 - Reaction time assessment
 - Sequence memory test
 - Symbol matching / processing speed test
- Designed and implemented the program interface layout:
 - Menu-based navigation system
 - Participant setup inputs
 - Structured test selection and execution flow
- Added participant profile inputs:
 - Age
 - Sex
 - Handedness
 - Sleep
 - Genetic background
- Connected the interface to motor test data collection:
 - Real-time tapping input capture
 - Cursor-based tremor tracking
 - Canvas-based tracing precision capture
 - Direct linkage between UI actions and backend computation
- Implemented a unified data integration layer:
 - Combined motor and cognitive test outputs
 - Added structured session-based storage
 - Added JSON export of full assessment results
 - Added feature vector generation for future model input

Notes and Adjustments

- Cognitive testing components were implemented earlier than initially scheduled.
 - The interface layer remains lightweight and mainly serves as a control and navigation system.
 - Additional cleanup was still needed:
 - Remove redundant NumPy import in the motor module
 - Continue refining input timing accuracy
-

Upcoming Phase: Dataset Integration and Model Development

Timeframe: April 8 – April 20, 2026

Planned Objectives

- Integrate external datasets such as PPMI and AMP-PD.
- Perform exploratory data analysis on reference datasets.
- Align PADM-generated features with dataset variables.
- Implement feature normalization and scaling.
- Begin similarity scoring using k-NN or distance-based comparison.

Future Additions

- Graph-based result visualization in the UI.
 - Similarity scoring display for non-diagnostic insight.
 - Preparation for real-world data collection and testing.
-

Program Launch Notes

Date: April 9, 2026

Running the Program

To run the PADM program:

1. Navigate to the project folder:

`/c/padm/PADM_Parkinson-sAnalysisThroughDigitalMotorMetrics`

2. Confirm the correct main file is present.
3. Run the program using:

`python padm_ui_layout.py`

4. The program window should open and begin running.
-

UI Touch-Up Specifications

General Test Instructions

- Add more specific instructions throughout all tests so participants understand what to do.

- Improve participant-facing clarity across the testing workflow.

Participant Setup

- Remove the manual Participant ID prompt.
- Automatically generate participant IDs internally.
- Specify clear options for the sex prompt:
 - Male
 - Female
 - Other
 - Prefer not to say

Tapping Test

- Fix spacebar tapping so input registers correctly.
- Add a moving timer to the UI.
- Expand the tapping test from 1 trial to 3 trials.
- Add a progress bar.
- Track the following metrics:
 - Total taps
 - Duration in seconds
 - Taps per second
 - Average interval in seconds
 - Interval standard deviation in seconds
- Average the trial results to produce participant-level tapping output.
- Return averaged data back into the workflow.

Tremor Rate Test

- Expand the tremor test from 1 trial to 3 trials.
- Add a progress bar.
- Fix the existing test frame so the tremor test becomes functional.
- Add a visible Start Trial button.
- Add a label to the handedness dropdown.
- Replace non_dominant with non-dominant.
- Add a note that an external touchpad is needed for best accuracy.

Tracing Test

- Add additional tracing tasks with different shapes.
- Add a progress bar.

- Track the following metrics:
 - Duration in seconds
 - Sample count
 - Mean deviation
 - Max deviation
 - Path efficiency
 - Completion ratio
- Average tracing results across tasks.
- Return averaged tracing data back into the workflow.

Results Tab

- Implement graphed results for each test.
 - Include:
 - Participant curve, labeled NPD
 - Parkinson's projected curve, labeled PD
-

Phase 2 Extension: UI Refinement and Motor Test Workflow Implementation

Timeframe: April 8 – April 14, 2026

Completed Objectives as of April 14, 2026

Participant-Facing UI

- Refined the participant-facing interface for a clearer testing workflow.
- Added more specific instructions across motor test screens.
- Removed the manual Participant ID prompt.
- Changed participant setup to automatically generate participant IDs internally.
- Added clear sex selection options:
 - Male
 - Female
 - Other
 - Prefer not to say

Tapping Test Improvements

- Fixed spacebar tapping registration within the UI workflow.
- Added a moving on-screen timer during tapping trials.

- Expanded tapping from 1 trial to 3 trials.
- Added a progress bar for tapping test completion.
- Captured per-trial tapping metrics:
 - Total taps
 - Duration in seconds
 - Taps per second
 - Average interval in seconds
 - Interval standard deviation in seconds
- Implemented averaging of tapping trial metrics.
- Returned averaged tapping data into the workflow and results pipeline.

Tremor Test Improvements

- Added a visible Start button to the tremor testing workflow.
- Added a label to the handedness / test-hand dropdown.
- Changed non_dominant display text to non-dominant.
- Added a participant-facing note about the external touchpad requirement.
- Expanded tremor testing from 1 trial to 3 trials.
- Added a progress bar for tremor test completion.
- Implemented an averaging structure for repeated tremor trial collection.
- Improved the tremor workflow so the existing frame is now functional and testable.

Tracing Test Improvements

- Added multiple tracing tasks with different shapes.
- Expanded tracing from 1 task to 3 tasks.
- Added a progress bar for tracing task completion.
- Captured per-task tracing metrics:
 - Duration in seconds
 - Sample count
 - Mean deviation
 - Max deviation
 - Path efficiency
 - Completion ratio
- Implemented averaging of tracing trial metrics.
- Returned averaged tracing data into the workflow and results pipeline.

Results Tab Improvements

- Implemented graph-based results displays for completed tests.
- Added participant curve visualization labeled NPD.

- Added Parkinson's projected curve visualization labeled PD.
- Connected repeated-trial motor results into a graph-ready output structure.

Notes and Adjustments

- Current PD and NPD graph curves are placeholder visual comparison curves for structure and UI testing.
 - These curves will later be replaced with dataset-driven baseline values after reference dataset integration is complete.
 - The motor test workflow now more closely reflects intended participant-facing use instead of backend-only scaffolding.
 - The program can still be launched through the main UI file.
-

Dataset Access Update

Date: April 22, 2026

PPMI Data Access

- PPMI data access application was approved.
 - This allows the project to begin exploring PPMI data for possible baseline comparison and reference-curve development.
-

PADM Program Adjustment Plan

Date: May 4, 2026

Menu Updates

- Neaten the main menu and make it look more professional.
- Use comforting colors.
- Add needed disclaimers.
- Fill empty space with a clearer explanation of the program's purpose.
- Add an About button on the home page.
- Link the About button to the project's GitHub.

Participant Setup Updates

- Add more detailed participant instructions.
- Keep participant setup simple and understandable.

Tapping Speed Test Updates

- Remove hand selection.
- Test three tapping modes:
 - Left hand
 - Right hand
 - Alternating tapping
- Each mode should be completed 3 times.
- Average the results across trials.
- Add more detailed instructions.
- Fix the issue where the results window disappears if the spacebar is accidentally pressed.
- Remove the results output window.
- Use the available screen space more effectively.

Tremor Testing Updates

- Add more detailed instructions.
- Fix the missing Start Trial button.
- Adjust the UI so it does not go off screen.
- Remove the results output window.
- Change the white capture area into a smaller, more visually appealing ball.

Tracing Test Updates

- Add more detailed instructions.
- Remove the results window at the bottom of the screen.
- Add a fourth tracing trial using a circular shape.

Cognitive Testing Updates

- Add more detailed instructions.
- Add a progress bar that moves as the participant continues through the cognitive tests.

Reaction Time Test

- Add more detailed instructions.
- Randomize wait time between 5 and 15 seconds before “GO” appears.
- Make the “GO” signal bigger and more visual.

- Make the result window more visually appealing.
- Display reaction time for each trial.
- Do not show raw output when “Save Reaction Result” is clicked.
- Use the available screen space more effectively.

Memory Test

- Add more detailed instructions.
- Prevent the user from typing while the sequence is being shown.
- Add a 5-second wait before the user can enter the number sequence.
- Add a trial number next to the memory progress bar, such as Trial: #/3.
- Remove the results output so the user does not know whether they were correct during testing.

Processing Speed Test

- Add more detailed instructions.
- Add a trial number next to the processing speed progress bar, such as Trial: #/3.
- Make the symbols larger.
- Change 2 of the trials from symbols to shapes.

Results Tab Updates

- Delete the visible raw results tab.
- Add a hidden key sequence to access raw results.
- Rename the motor graph tab to:

Motorskill Results

- Implement PPMI into graphs where appropriate.
- Add correlation and reference-curve structure.
- Make graphs more visually appealing.

Cognitive Graph Updates

- Rename the cognitive graph tab to:

Cognitive Test Results

- Improve the graph appearance.
- Find a PPMI or related dataset that can correlate with cognitive testing.
- Build a PD curve where data allows.

Additional Notes

- Autosave all cognitive testing results as the user progresses.
 - Continue making the entire program look professional, neat, and participant-friendly.
-

PADM Change Log

Date: May 5, 2026

File Changed: padm_ui_layout.py

- Replaced the previous UI file with an updated participant-facing interface.
- Refined the menu and home layout to look cleaner, more professional, and more comforting.
- Added clearer project-purpose language on the home screen.
- Added non-diagnostic disclaimer language explaining that PADM provides comparative insight, not a medical diagnosis.
- Improved participant setup instructions.
- Preserved automatic participant ID generation instead of manual participant ID entry.

Tapping Test Updates

- Removed the hand-selection dropdown.
- Added left-hand tapping, right-hand tapping, and alternating tapping modes.
- Structured each tapping mode to run 3 trials.
- Added averaging logic for repeated tapping trials.
- Reduced or removed raw output display during the participant test flow.
- Improved spacebar input handling.

Tremor Test Updates

- Added and fixed the visible Start Trial button.
- Added clearer tremor test instructions.
- Added an external touchpad note for best accuracy.
- Removed the raw results/output window from the test screen.
- Reworked the capture area to be smaller and more visually controlled.
- Preserved the 3-trial tremor workflow and averaged result structure.

Tracing Test Updates

- Added clearer tracing instructions.
- Removed the raw output window from the test screen.
- Added a fourth tracing trial using a circular shape.
- Preserved averaged tracing result output.

Cognitive Testing Updates

- Added cognitive progress tracking.
- Added autosave behavior as cognitive tests are completed.
- Updated the reaction time test with a randomized 5–15 second wait before the GO signal.
- Enlarged and emphasized the GO signal.
- Updated the memory test so users cannot type while the sequence is being shown.
- Added a 5-second wait before memory sequence entry.
- Added trial labels such as Trial: #/3.
- Removed participant-facing correctness feedback during memory testing.
- Updated the processing speed test with larger symbols and shapes.

Results Section Updates

- Removed the visible raw-results tab from normal participant view.
 - Added hidden raw-results access through Ctrl + Shift + R.
 - Renamed result sections to:
 - Motorskill Results
 - Cognitive Test Results
 - Improved graph display structure for motor and cognitive outputs.
 - Integrated the PPMI mobility reference curve into the Motorskill Results tab.
 - Added automatic loading of ppmi_mobility_curve.py from the project folder.
 - Added automatic search for the Neuro-QoL mobility CSV in the project folder.
 - Added the PPMI Neuro-QoL mobility impairment trend graph using EVENT_ID visit points.
 - Added dataset summary text showing:
 - Valid visit records
 - Unique participants
 - Mean mobility impairment score
 - Updated the results notice to explain that the PPMI graph is mobility-based and does not directly validate tapping, tremor, cursor movement, or tracing metrics.
-

File Changed: padm_motor_tests.py

- Updated backend motor testing support for the refined UI workflow.
- Added cleaner label formatting for internal names such as non_dominant.

Tapping Backend Updates

- Added support for:
 - Left-hand tapping
 - Right-hand tapping
 - Alternating tapping
 - Fixed-duration scoring
 - Left and right tap counts
 - Alternation accuracy
 - Battery-level averaging across tapping modes

Tremor Backend Updates

- Preserved FFT-based tremor frequency estimation when NumPy is available.
- Preserved fallback tremor estimation when NumPy is unavailable.
- Added movement-area calculation.
- Added signal-quality labels:
 - Usable
 - Low sample rate
 - Low motion
 - Insufficient data
- Added averaged tremor frequency standard deviation across trials.

Tracing Backend Updates

- Added shape_name tracking.
- Added circular tracing path helper.
- Added target path length and traced path length.
- Improved completion-ratio calculation.
- Improved circular path-efficiency handling.
- Improved averaging helper functions for tapping, tremor, and tracing outputs.

File Changed: padmCognitiveTests.py

- Updated the cognitive backend to better support the refined UI workflow.

- Preserved a backend-only structure so the UI controls participant-facing display.

Reaction Time Updates

- Expanded reaction time result storage:
 - Average reaction time
 - Minimum reaction time
 - Maximum reaction time
 - Standard deviation
 - Individual trial reaction times

Sequence Memory Updates

- Improved sequence memory result storage:
 - Total rounds
 - Correct rounds
 - Longest correct span
 - Accuracy
 - Attempted spans
 - Round-level details
- Added support for hiding memory correctness from the participant while still saving correctness internally.

Processing Speed Updates

- Preserved the original record_response(is_correct) method.
- Added detailed item-level recording support.
- Added support for symbol-based and shape-based trials.
- Tracked symbol items and shape items separately.
- Stored item-level details for future graphing and hidden raw-results review.

Cognitive Session Wrapper Updates

- Saves result objects and dictionary-style results.
- Tracks completed cognitive test sections.
- Supports autosaving cognitive results as the participant progresses.

File Changed: padm_data_integration.py

- Strengthened the central data integration layer for the updated UI workflow.

- Added updated_at tracking to assessment records.
 - Added optional autosave support.
 - Updated motor-result saving to support autosave after completed motor tests.
 - Updated cognitive-result saving to support autosave after completed cognitive tests.
 - Added cleaner result-name formatting for export and graphing.
 - Improved JSON export support for hidden raw-results access.
 - Improved feature-vector generation for future similarity scoring and dataset comparison.
 - Updated feature-vector logic to keep scalar and model-friendly values while skipping raw-heavy fields such as trial lists and item-level details.
 - Added model_ready_features() for future scoring and model use.
 - Added ppmi_alignment_template() as a future mapping structure for PPMI and AMP-PD comparison.
-

File Added: ppmi_mobility_curve.py

- Added a separate PPMI mobility reference-curve module.
 - Loads the Neuro-QoL Lower Extremity Function / Mobility short-form CSV.
 - Calculates raw mobility score using 8 Neuro-QoL mobility item columns:
 - NQMOB37
 - NQMOB30
 - NQMOB26
 - NQMOB32
 - NQMOB25
 - NQMOB33
 - NQMOB31
 - NQMOB28
 - Converts raw mobility score into a 0–100 mobility impairment score.
 - Builds an event-based PPMI mobility reference curve using EVENT_ID.
 - Generates dataset summary statistics.
 - Exports the reference curve to JSON.
 - Can generate a standalone mobility reference graph image.
 - Establishes the first dataset-driven PD reference curve in the project.
 - Notes that the curve reflects general lower-extremity mobility impairment, not direct validation of tapping, tremor, cursor movement, or tracing metrics.
-

File Changed: PADM_MATH_AND_DATA_FLOW.md

- Updated documentation to reflect the revised PADM workflow.
 - Documented the updated participant-facing UI structure.
 - Documented the revised motor testing workflow.
 - Documented the revised cognitive testing workflow.
 - Documented hidden raw-results access.
 - Documented PPMI mobility reference-curve integration.
 - Clarified that current PPMI integration is mobility-based and not yet a direct PADM feature correlation.
 - Added notes for future dataset mapping, similarity scoring, and direct motor-feature comparison.
-

Dataset Added to Project Folder

File Added:

Neuro_QoL__Lower_Extremity_Function__Mobility__ - _Short_Form_22Apr2026.csv

- Added the approved PPMI Neuro-QoL Lower Extremity Function / Mobility short-form CSV to the project folder.
 - Used as the first dataset source for generating a mobility-based PD reference curve.
 - Supports event-based mobility trend graphing through EVENT_ID.
 - Used by ppmi_mobility_curve.py.
 - Displayed in the UI through padm_ui_layout.py.
-

Current Implementation Status

- UI refinement: Implemented.
- Motor test workflow refinement: Implemented.
- Cognitive test workflow refinement: Implemented.
- Result graph restructuring: Implemented.
- Hidden raw-results access: Implemented.
- PPMI mobility reference curve: Implemented.